「Thm. Let $\{a_n\}$ be a sequence of real numbers, and let $S_k \overset{\text{def}}{=} a_1 + \cdots + a_k$ ($k \in \mathbb{N}$).
If $\lim_{n\to\infty} a_n = A \in \mathbb{R}$, then $\displaystyle\lim_{n\to\infty} \frac{S_n}{n} = A$.

Proof. Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow |a_n - A| < \varepsilon$
Meanwhile, observe that

$$\left|\frac{S_n}{n} - A\right| = \frac{1}{n}\cdot\left|a_1 + \cdots + a_n - nA\right| = \frac{1}{n}\cdot\left|a_1 - A + a_2 - A + \cdots + a_n - A\right|$$

Now, for all $n \geq N$,

$$\begin{aligned}
\left|\frac{S_n}{n} - A\right| &= \frac{1}{n}\cdot\left|a_1 - A + \cdots + a_N - A + a_{N+1} - A + \cdots + a_n - A\right| \\
&\leq \frac{1}{n}\cdot\left(|a_1 - A| + |a_2 - A| + \cdots + |a_n - A|\right) \\
&= \frac{1}{n}\cdot\left(\sum_{i=1}^{N}|a_i - A|\right) + \frac{1}{n}\cdot\left(\sum_{i=N+1}^{n}|a_i - A|\right) \\
&\leq \frac{1}{n}\cdot\left(\sum_{i=1}^{N}|a_i - A|\right) + \frac{1}{n}\cdot(n-N)\cdot\varepsilon \\
&\leq \frac{1}{n}\cdot\left(\sum_{i=1}^{N}|a_i - A|\right) + \varepsilon
\end{aligned}$$

Since N was fixed, $\displaystyle\limsup_{n\to\infty}\left|\frac{S_n}{n} - A\right| \leq \limsup_{n\to\infty}\frac{1}{n}\left(\sum_{i=1}^{N}|a_i - A|\right) + \varepsilon = \varepsilon$.

Since $\varepsilon > 0$ was arbitrary, $\displaystyle\limsup_{n\to\infty}\left|\frac{S_n}{n} - A\right| = 0$, so $\displaystyle\lim_{n\to\infty}\frac{S_n}{n} = A$. ∎

This theorem is called the Cesaro mean,
and in the next page, we deal with the generalization which is called the Stolz-Cesaro theorem.

「Stolz-Cesaro Theorem.

Let $\{a_n\}$ be a sequence of real numbers,
and $\{b_n\}$ be a strictly increasing which diverges to $+\infty$.

Then, $$\liminf_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n} \le \liminf_{n\to\infty} \frac{a_n}{b_n} \le \limsup_{n\to\infty} \frac{a_n}{b_n} \le \limsup_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n}.$$

Moreover, if the limit $\lim_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n}$ exists,

then $$\lim_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n} = \lim_{n\to\infty} \frac{a_n}{b_n}$$